

# PIA 407 Course Outline

## PIA 407 – Risk, Sensitivity and Switching Values

*The number at which the recommendation changes.*

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|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>School</b>            | 04 – Public Investment                                                                                                                                                 |
| <b>Hours</b>             | 30 (indicative band 27-33)                                                                                                                                             |
| <b>Version</b>           | 1.0 (2026-09-05)                                                                                                                                                       |
| <b>Model</b>             | AICTEM PIAS,<br>D:/EDIP/models/PIASModelling                                                                                                                           |
| <b>Reference country</b> | Utopia (UTP), UTD 2024 prices                                                                                                                                          |
| <b>Prerequisites</b>     | PIA 404, whose ENPV every test in this course perturbs. PIA 403 for discounting. PIA 406 is useful, because the conversion factors are among the things worth testing. |
| <b>Audience</b>          | Appraisal economists, reviewers, and the officials who have to decide how much confidence a point estimate deserves                                                    |
| <b>Pass mark</b>         | 60 %                                                                                                                                                                   |

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### What this course is

An appraisal produces one number. The number is a function of several dozen assumptions, some of which are governed parameters, some sector-model outputs, and some the analyst's own judgement. The question this course answers is not whether the number is right, which nobody can know, but how far any of those assumptions would have to move before the recommendation changed.

It teaches three things in order. One-way sensitivity, which perturbs a single variable and reports the elasticity of the result. Switching values, which find the point at which the ENPV reaches zero – by bisection on the full appraisal recomputation, not by extrapolating from a sensitivity table, because ENPV is not linear in a delay, in the discount rate, or in cost once contingency and optimism bias interact. And the risk register, which is the qualitative counterpart: what could go wrong, how likely, how bad, and who owns it.

The register supplies 143 sensitivity results across 13 standard tests, of which 23 switch the sign of the ENPV, and 55 switching values – including several the engine refuses to report, with the reason stated.

Its central discipline is that a refusal is a result. Where the base case is already negative the engine returns no switching value and says: “the base case is already negative, so there is no switching value to find - the question is what would make the project viable, not what would make it unviable” That sentence is more useful than any number it could have produced.

## What a graduate can do

Run a one-way sensitivity suite and read its elasticities; identify the variables to which a result is most and least sensitive; compute a switching value by bisection on the full recomputation and explain why linear extrapolation from a sensitivity table is unsafe; identify the cases in which no switching value exists and state the engine's reason; build a combined downside case and say why it is not the sum of the individual tests; construct a risk register with probability, impact, exposure, mitigation and an owner; distinguish inherent from residual risk; and write a risk section that reports the margin rather than the point estimate.

## What this course does not claim

This course does not put a distribution round the answer. One-way sensitivity moves one variable at a time and a switching value is a single point; neither says how likely any of it is. PIA 408 does that, with Monte Carlo simulation, correlation and a reported probability of a negative ENPV.

It does not estimate probabilities. The register's probability scale is a five-point ordinal scale with attached numeric values, and those values are a convention rather than an estimate. Where a genuine probability is required the evidence must come from somewhere, and this course says where rather than supplying one.

It does not price risk. A switching value tells you the margin; it does not tell you what that margin is worth. Risk pricing belongs to the contract – who bears the cost overrun – and to the contingent liability accounting in a PPP, neither of which is covered here.

It does not test the conversion factors. Testing a governed parameter is legitimate and this course explains how, but the register's stored sensitivity suite perturbs cost, benefit, demand, operating cost, delay and the discount rate – not the shadow prices.

## Learning outcomes

1. State what a sensitivity analysis is for, and what it cannot tell you.
2. Run a one-way sensitivity suite and read the resulting elasticities.
3. Rank variables by the sensitivity of the result to them.
4. Define a switching value and say why it is found by bisection on the full recomputation.
5. Explain why linear extrapolation from a sensitivity table is unsafe.
6. Compute a switching value and express it as a margin.
7. Identify the cases in which no switching value exists, and quote the engine's reason.
8. Build a combined downside case and say why it is not the sum of the individual tests.
9. Construct a risk register with probability, impact, exposure, mitigation and an owner.
10. Distinguish inherent from residual risk.
11. Say why an exposure score is a ranking device rather than an expected value.
12. Write a risk section that reports the margin rather than the point estimate.

## Module structure

| Module | Title                   | Sessions | Hours | Laboratory | Manual | Assessment |
|--------|-------------------------|----------|-------|------------|--------|------------|
| M01    | What sensitivity is for | 1        | 1.25  | –          | 1      | Quiz M01   |

| Module | Title                            | Sessions | Hours | Laboratory | Manual | Assessment |
|--------|----------------------------------|----------|-------|------------|--------|------------|
| M02    | The one-way suite                | 1        | 3.25  | Lab 01     | 2      | Quiz M02   |
| M03    | Elasticity and ranking           | 1        | 1.50  | –          | 3      | Quiz M03   |
| M04    | The switching value              | 1        | 3.50  | Lab 02     | 4      | Quiz M04   |
| M05    | Why bisection, not extrapolation | 1        | 1.50  | –          | 5      | Quiz M05   |
| M06    | Margins, and how to report one   | 1        | 3.50  | Lab 03     | 6      | Quiz M06   |
| M07    | When there is no switching value | 1        | 1.50  | –          | 7      | Quiz M07   |
| M08    | The combined downside            | 1        | 1.50  | –          | 8      | Quiz M08   |
| M09    | The risk register                | 1        | 3.50  | Lab 04     | 9      | Quiz M09   |
| M10    | Probability, impact and exposure | 1        | 1.50  | –          | 10     | Quiz M10   |
| M11    | Inherent, residual and the owner | 1        | 3.50  | Lab 05     | 11     | Quiz M11   |
| M12    | Writing the risk section         | 1        | 1.50  | –          | 12     | Quiz M12   |

## Laboratories

| Lab | Title                                                                | Hours | Deliverable                               |
|-----|----------------------------------------------------------------------|-------|-------------------------------------------|
| 1   | Run the one-way suite and rank the variables                         | 2.0   | A ranked sensitivity table                |
| 2   | Compute switching values by bisection, and read the convergence flag | 2.0   | A verified switching value table          |
| 3   | Turn switching values into margins, and benchmark them               | 2.0   | A margins table and a reporting paragraph |

| Lab | Title                                                                     | Hours | Deliverable                     |
|-----|---------------------------------------------------------------------------|-------|---------------------------------|
| 4   | Build a risk register, and score it honestly                              | 2.0   | A scored risk register          |
| 5   | Allocate each risk, cost the mitigation, and trace it to the cost profile | 2.0   | A costed and allocated register |

## Assessment

| Component                | Weight |
|--------------------------|--------|
| Knowledge quizzes        | 15 %   |
| Laboratory exercises     | 30 %   |
| Applied assignments      | 25 %   |
| Capstone / final project | 30 %   |

## Applied assignments

| ID | Title                                       | Weight | Deliverable                                                           |
|----|---------------------------------------------|--------|-----------------------------------------------------------------------|
| A1 | The suite and its duplicate lever           | 25     | Marked on the ranking and on detecting that two rows are one          |
| A2 | Switching values, computed and extrapolated | 25     | Marked on the verification and on the discount rate contrast          |
| A3 | The combined downside                       | 25     | Marked on the structural reasons, not the arithmetic alone            |
| A4 | The register, allocated and costed          | 25     | Marked on the allocation reasoning and on both sides of the trade-off |

## Final project

A complete risk and sensitivity section for one project: the one-way suite with elasticities and the variables ranked by sensitivity; every switching value with its margin, and for each that could not be computed the engine's reason quoted in full; a combined downside case with a statement of why it is not the sum of the parts; a risk register with probability, impact, exposure, mitigation, residual exposure and a named owner for each entry; and a closing paragraph that reports the margin rather than the point estimate and says what would change the recommendation.

## Folder contents

Every course in the Academy carries the same section 5 folder. See `README.md` for what each item holds and `COURSE_BUILD_REPORT.md` for what was measured.